Discriminative Project Milestone #4

Group 5

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GitHub

1. Project Overview

This project builds a deep learning system that can identify a single object from an image and detect multiple objects in a composite image. The system supports two tasks:

- Single-object image identification
- Multi-object detection with object ID and bounding box prediction

The dataset was created by students in the course. For single-object classification, the dataset contains 4,108 RGB images across 39 object classes . The data was split using stratified sampling into:

- 2,875 training images
- 616 validation images
- 617 test images

Three CNN models were trained for single-object classification:

- SimpleCNN (custom model trained from scratch)
- ResNet18 (pretrained on ImageNet)
- MobileNetV3Small (pretrained on ImageNet)

Final test performance for single-object classification:

- ResNet18: 98.87% test accuracy, 98.84% macro F1
- MobileNetV3Small: 96.76% test accuracy
- SimpleCNN: 60.45% test accuracy

ResNet18 was selected as the best single-object classification model based on validation and test performance.

For multi-object detection, a grid-based composite dataset was generated by combining single-object images into multi-object images . The detection dataset contains:

- 100 training images
- 20 validation images
- 39 object classes

YOLOv8 was used for object detection with transfer learning . Multiple variants were trained and compared:

- YOLOv8 Nano
- YOLOv8 Small
- YOLOv8 Medium
- YOLOv8 Nano (Optimized)

Validation performance comparison:

- YOLOv8 Nano: mAP@0.5 = 0.8269, mAP@0.5:0.95 = 0.8224, Precision = 0.7665, Recall = 0.7361
- YOLOv8 Small: mAP@0.5 = 0.9868, mAP@0.5:0.95 = 0.9838, Precision = 0.9559, Recall = 0.9537
- YOLOv8 Medium: mAP@0.5 = 0.9853, mAP@0.5:0.95 = 0.9836, Precision = 0.9499, Recall = 0.9609
- YOLOv8 Nano (Optimized): mAP@0.5 = 0.9550, mAP@0.5:0.95 = 0.9516

YOLOv8 Small achieved the highest overall detection performance and was selected for deployment.

For single-object detection using YOLO inference, the trained model returns:

- Object ID (class index)
- Class name (e.g., OBJ029)
- Confidence score (example: 93.37%)
- Bounding box coordinates

This confirms that the system can identify object ID and location for both single-object and multi-object images using trained CNN and YOLOv8 models.

2. Dataset Details

2.1 Single-Object Classification Dataset

The single-object dataset was created by students in the course. Each student contributed images of one object under different lighting conditions and angles.

Dataset statistics:

- Total images: 4,108
- Number of classes: 39
- Images per class: approximately 100–110

The dataset was split using stratified sampling to maintain equal class distribution:

- Training set: 2,875 images
- Validation set: 616 images
- Test set: 617 images

Preprocessing steps:

- All images resized to 160 × 160 pixels
- Normalization using ImageNet means and standard deviation
- Data augmentation applied only to training set:
    - Random horizontal flip
    - Random rotation (±8 degrees)
    - Color jitter (±12%)

Images were converted to PyTorch tensors before training.

2.2 Multi-Object Detection Dataset

The multi-object dataset was generated by combining single-object images into grid-based composite images. Each composite image contains multiple objects placed at known locations.

Dataset statistics:

- Total multi-object images: 120
- Training images: 100
- Validation images: 20
- Number of classes: 39

The dataset follows the YOLOv8 directory structure with:

- images/train
- images/val
- labels/train
- labels/val
- data.yaml configuration file

Each object in an image has a corresponding annotation in YOLO format:

- Class ID
- Normalized center x
- Normalized center y
- Width
- Height

The data.yaml file defines:

- Dataset path
- Training and validation directories
- Number of classes
- Class names in OBJ format (e.g., OBJ001, OBJ029)

Image resizing and augmentation for detection were handled internally by the YOLOv8 training pipeline.

3. End-to-End Workflow and Process

This section describes the full pipeline from data preparation to final inference.

Raw Object (Physical)
↓
[M1] Image Collection & Preprocessing
↓
[M1] Single-Object Model Training & Evaluation
↓
[M2] Multi-Object Synthetic Dataset Generation (PIL compositing)
↓
[M2] YOLO Annotation (auto bounding box → .txt labels)
↓
[M3] Dataset Split + data.yaml config
↓
[M3] YOLOv8 Custom Training (Google Colab, T4 GPU)
↓
[M3] Evaluation (mAP, Precision, Recall, F1)
↓
[M4] Demo Results & Final Report

3.1 Single-Object Classification Workflow

Step 1: Data Preparation

- Load 4,108 RGB images across 39 classes
- Apply stratified split into train (2,875), validation (616), and test (617) sets
- Resize images to 160 × 160
- Apply normalization using ImageNet statistics
- Apply augmentation on training data only

Step 2: Model Training

Three models were trained under identical conditions:

- SimpleCNN (custom CNN)
- ResNet18 (pretrained)
- MobileNetV3Small (pretrained)

Training configuration:

- Optimizer: Adam
- Learning rate: 0.0001
- Batch size: 32
- Epochs: 3

Step 3: Model Evaluation
Evaluation Metrics:

- Test accuracy
- Macro F1 score
- Confusion matrix

Final test performance:

- ResNet18: 98.87% accuracy, 98.84% macro F1
- MobileNetV3Small: 96.76% accuracy
- SimpleCNN: 60.45% accuracy

ResNet18 weights were saved and used as the final single-object classification model.

3.2 Multi-Object Detection Workflow

Step 1: Multi-Object Image Generation

- Combine single-object images into grid-based composite images
- Generate YOLO-format bounding box annotations
- Verify dataset structure and counts (100 train, 20 validation)

Step 2: Model Training with Transfer Learning
YOLOv8 pretrained weights were used as initialization .
Training configuration:

- Image size: 640
- Batch size: 16
- Epochs: 50
- Early stopping with patience
- Automatic mixed precision enabled

Models trained:

- YOLOv8 Nano
- YOLOv8 Small
- YOLOv8 Medium
- YOLOv8 Nano (Optimized with augmentation and AdamW)

Step 3: Model Evaluation
Metrics extracted after validation:

- mAP@0.5
- mAP@0.5:0.95
- Precision
- Recall

Validation performance:

- YOLOv8 Small: mAP@0.5 = 0.9868, mAP@0.5:0.95 = 0.9838, Precision = 0.9559, Recall = 0.9537
- YOLOv8 Medium: mAP@0.5 = 0.9853, mAP@0.5:0.95 = 0.9836
- YOLOv8 Nano: mAP@0.5 = 0.8269

YOLOv8 Small was selected as the final detection model.

4. Model Architecture and Training

4.1 Single-Object Classification Models

All models were modified for 39-class classification.

SimpleCNN

- 4 convolution layers
- Conv → ReLU → MaxPool repeated 4 times
- Fully connected layer with dropout (0.3)
- Final linear layer outputs 39 classes
- Trained from scratch without pretrained weights
- Approximately 11 million parameters

ResNet18

- 18-layer residual network
- Pretrained on ImageNet
- Final fully connected layer replaced with 39-class output layer

MobileNetV3Small

- Lightweight CNN architecture
- Pretrained on ImageNet
- Final classifier layer modified for 39 classes

Training setup for all three models:

- Optimizer: Adam
- Learning rate: 0.0001
- Batch size: 32
- Epochs: 3
- Loss function: CrossEntropyLoss
- Device: CPU

Best model weights were saved based on validation accuracy.

4.2 Multi-Object Detection Model (YOLOv8)

YOLOv8 uses a single-stage detection architecture where classification and bounding box regression are performed in one forward pass .

Model parameter counts:

- YOLOv8 Nano: 758,917 parameters
- YOLOv8 Small: 2,131,141 parameters
- YOLOv8 Medium: 3,798,277 parameters

Training configuration:

- Image size: 640
- Batch size: 16
- Epochs: 50
- Early stopping with patience = 10
- Automatic mixed precision enabled
- Pretrained weights loaded before fine-tuning

For the optimized Nano model:

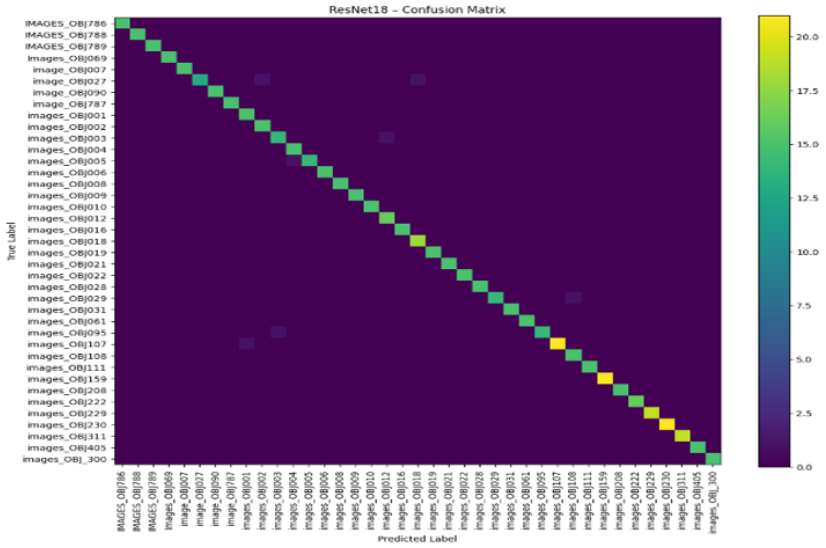
- Data augmentation included rotation, translation, scaling, horizontal flip, mosaic
- Optimizer changed to AdamW
- Learning rate adjusted
- Patience increased

5. Model Testing and Evaluation

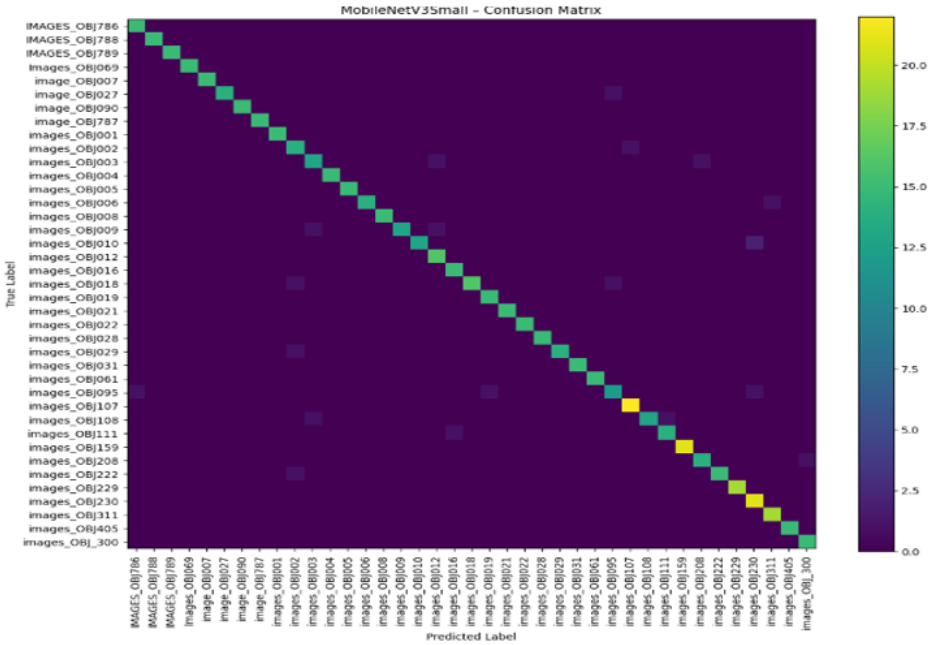
5.1 Single-Object Classification Evaluation

The single-object classification models were evaluated on a held-out test set of 617 images. Performance was measured using test accuracy and macro F1 score.
ResNet18 achieved a test accuracy of 98.87% and a macro F1 score of 98.84%. MobileNetV3Small achieved a test accuracy of 96.76%. SimpleCNN achieved a test accuracy of 60.45% and a macro F1 score of 57.66%.
Validation accuracy during training reached 99.7% for ResNet18 by epoch 3. MobileNetV3Small reached 97.2% validation accuracy by epoch 3. SimpleCNN reached 62.0% validation accuracy by epoch 3.

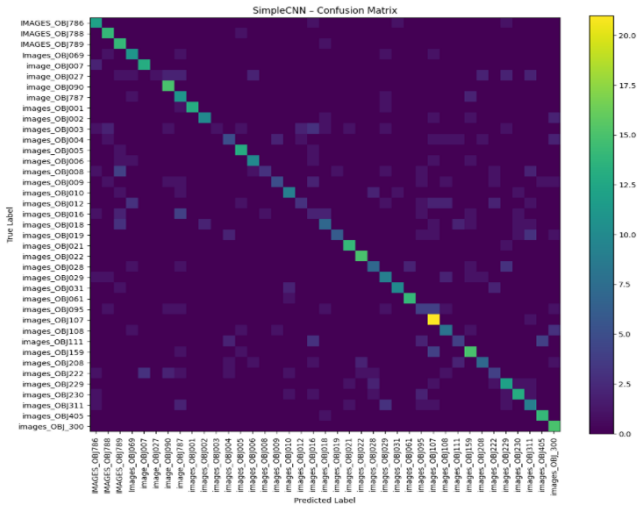
Confusion matrix for ResNet18



Confusion matrix for MobileNetV3Small



Confusion matrix for SimpleCNN



These matrices show prediction distribution across the 39 object classes.

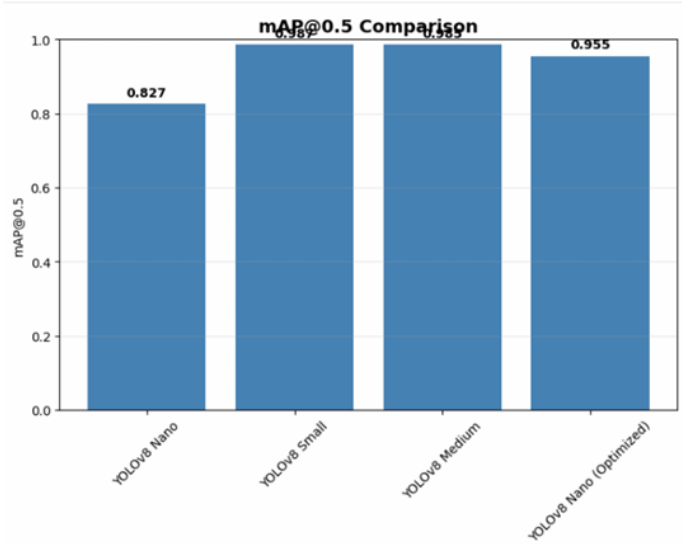
5.2 Multi-Object Detection Evaluation

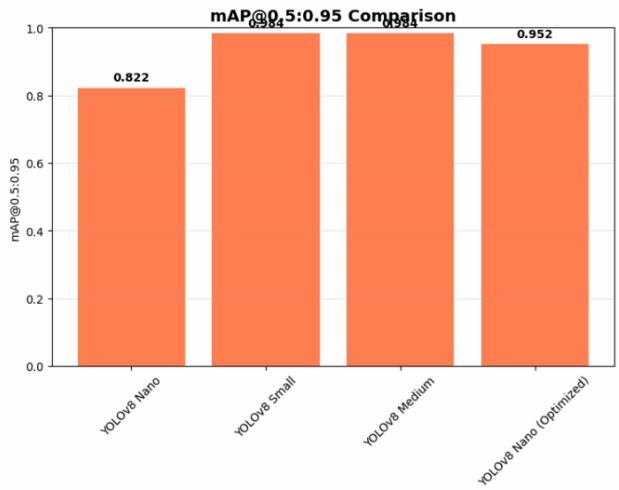
The multi-object detection models were evaluated on 20 validation images. The evaluation metrics included mAP@0.5, mAP@0.5:0.95, precision, and recall.

Model comparison table

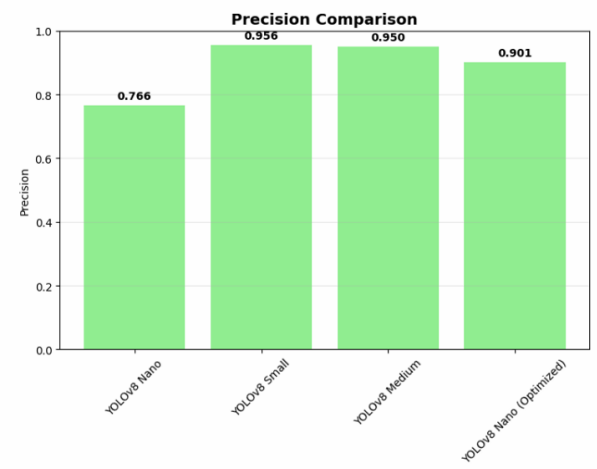
Model Name	mAP@0.5	mAP@0.5:0.95	Precision	Recall	Parameters
YOLOv8 Nano	0.8269	0.8224	0.7665	0.7361	758,917
YOLOv8 Small	0.9868	0.9838	0.9559	0.9537	2,131,141
YOLOv8 Medium	0.9853	0.9836	0.9499	0.9609	3,798,277
YOLOv8 Nano (Optimized)	0.9550	0.9516	0.9012	0.8826	758,917

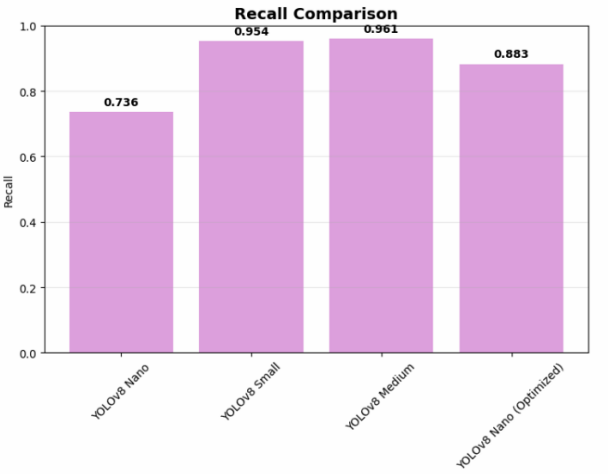
mAP comparison plot





Precision and recall comparison plots from the Milestone 2 report





The validation results show that YOLOv8 Small achieved the highest mAP@0.5 and mAP@0.5:0.95 among the tested variants.

6. Demo Results

This section presents inference outputs from the trained single-object classification model and the trained YOLOv8 detection model.

6.1 Single-Object Classification Demo

For single-object classification, an input image is resized to 160 × 160, normalized, and passed through the trained ResNet18 model . The model outputs a probability distribution over 39 object classes, and the predicted class corresponds to the highest probability.

```
----------------------------
Classification report: ResNet18
============================
                 precision    recall  f1-score   support

 IMAGES_OBJ786       1.00      1.00      1.00        15
 IMAGES_OBJ788       1.00      1.00      1.00        15
```

IMAGES_OBJ789	1.00	1.00	1.00	15
Images_OBJ069	1.00	1.00	1.00	15
image_OBJ007	1.00	1.00	1.00	15
image_OBJ027	1.00	0.87	0.93	15
image_OBJ090	1.00	1.00	1.00	15
image_OBJ787	1.00	1.00	1.00	15
images_OBJ001	0.94	1.00	0.97	15
images_OBJ002	0.94	1.00	0.97	15
images_OBJ003	0.93	0.93	0.93	15
images_OBJ004	0.94	1.00	0.97	15
images_OBJ005	1.00	0.93	0.97	15
images_OBJ006	1.00	1.00	1.00	15
images_OBJ008	1.00	1.00	1.00	15
images_OBJ009	1.00	1.00	1.00	15
images_OBJ010	1.00	1.00	1.00	15
images_OBJ012	0.94	1.00	0.97	16
images_OBJ016	1.00	1.00	1.00	15
images_OBJ018	0.95	1.00	0.97	18
images_OBJ019	1.00	1.00	1.00	15
images_OBJ021	1.00	1.00	1.00	15
images_OBJ022	1.00	1.00	1.00	15
images_OBJ028	1.00	1.00	1.00	15
images_OBJ029	1.00	0.93	0.97	15
images_OBJ031	1.00	1.00	1.00	15
images_OBJ061	1.00	1.00	1.00	15
images_OBJ095	1.00	0.93	0.97	15
images_OBJ107	1.00	0.95	0.98	22
images_OBJ108	0.94	1.00	0.97	15
images_OBJ111	1.00	1.00	1.00	15
images_OBJ159	1.00	1.00	1.00	21
images_OBJ208	1.00	1.00	1.00	15
images_OBJ222	1.00	1.00	1.00	16
images_OBJ229	1.00	1.00	1.00	19
images_OBJ230	1.00	1.00	1.00	21
images_OBJ311	1.00	1.00	1.00	19
images_OBJ405	1.00	1.00	1.00	15
images_OBJ_300	1.00	1.00	1.00	15
accuracy			0.99	617
macro avg	0.99	0.99	0.99	617
weighted avg	0.99	0.99	0.99	617

6.2 Single-Object Detection using YOLO

For YOLO inference on a single-object image, the trained detection model is loaded and applied with a confidence threshold of 0.25. The output includes **(Demo Image)**:

From the provided inference example:

- Object ID: 24
- Class: OBJ029
- Confidence: 93.37%

This demonstrates that the detection model can localize and identify a single object in an image.

6.3 Multi-Object Detection Demo

For multi-object inference, a composite image containing multiple objects is passed through the trained YOLOv8 Small model. The model outputs multiple bounding boxes**(Demo Image)**, each with:

- Class ID
- Confidence score
- Bounding box location

These outputs show that the model detects and assigns IDs to multiple objects in a single image.

7. Conclusion

This project implemented a full pipeline for single-object classification and multi-object detection using deep learning. The system was trained on a custom dataset of 39 object classes and evaluated using separate validation and test splits.

For single-object classification, pretrained CNN models achieved high test performance, with ResNet18 reaching 98.87% accuracy. For multi-object detection, transfer learning with YOLOv8 enabled accurate object ID prediction and bounding box localization, with YOLOv8 Small achieving mAP@0.5 of 0.9868 on the validation set.

The final system supports both identification of a single object and detection of multiple objects in a composite image using trained models and saved weights.